

Can a cheap EEG headset test what people report hearing?

A pre-registered, blinded, three-group EEG study of reported "voice-to-skull" experiences — three-page proposal for a prospective academic partner

Prepared for Dr O. Valentin · Stephane van der Aa, TI One Voice, Brussels, Belgium · 29 August 2026

1. Context and background

The population. A large, internationally distributed group of people — self-described "targeted individuals" (TIs) — report hearing voices they believe are technologically transmitted, a phenomenon they call voice-to-skull (V2K). Online communities run to hundreds of thousands of accounts; no reliable prevalence figure exists. Many report frequent, conversational voices perceived inside the head.

The mainstream account. Psychiatry reads these reports as auditory verbal hallucinations (AVH) within a persecutory delusional system. That account is coherent and well supported: AVH content matches, the technological attribution follows the era, and the source-monitoring model rests on a finding replicated for twenty-five years. We assume it is the most probable explanation for most of this population.

The gap. It has never been tested with instruments. The best-known formal study (Sheridan & James, 2015) is a narrative analysis of self-report; there is no instrumented study of this group at all, and the corollary discharge paradigm — applied extensively to clinical AVH — has never been applied to it. "Most probable" and "tested" are different states. This proposal closes that gap with a study designed so that either outcome is publishable and neither side can dismiss the method after the fact.

What it does not do. An EEG amplifier is a voltmeter on the scalp. It cannot detect a transmission, decode words, localise a source with eight channels, or tell an individual anything about themselves. The study measures brain state and self-monitoring function. No claim about the origin of the phenomenon is made anywhere in the design, and none is requested of a partner.

2. Methodology

Design. Three groups of ~25 (26/group gives 80% power at $d = 0.8$, the effect size in the AVH literature): (i) people reporting frequent V2K voices; (ii) no-voice controls matched on age, sex and handedness; (iii) a clinical AVH comparison group with an established explanation for their voices. Group (iii) is what makes the study decisive rather than suggestive, and it is the one that requires an academic or clinical partner. One ~2-hour session at home, six blocks with two breaks, the primary block repeated at the end as a within-session reliability check; participants who opt in to individual feedback complete three sessions a week apart, with the kit staying with them.

Four tests, one session.

- **Test 1 (primary) — self-generated sound discount.** TALK block: ~100 self-paced "ah" vocalisations, recorded. LISTEN block: level-matched playback of that recording. Measure N100 at Cz (80–130 ms) in both, subtract. Order counterbalanced. Reduced suppression in the study group is the pattern predicted by the internal-generation account.
- **Test 2 — auditory cortex state during reported episodes.** Random 1000 Hz probe tones every 3–8 s while the participant holds a button during reported voices; compare probe N100 inside vs outside reported periods with guard windows around presses. A provocation variant (scripted utterance → per-participant guard interval → post-speech window) contrasts reply vs no-reply trials within condition and yields a reply-latency distribution, with matched read-aloud and address-a-recording conditions to control expectancy.
- **Test 3 — silent reply.** Conditions: A silent recitation, B silent reply to a recorded question (timestamped ground truth), C silent reply to the reported voice. Four channels: EEG, speech-muscle EMG (subvocalisation), readiness potential, and turn-taking structure tested by permutation. Cross-condition decoding: train A vs B (regularised linear model on tangent-space covariance and ERP features, subject-wise nested CV, empirical chance by label shuffling, calibrated probabilities), apply to C, never seen in training.
- **Test 4 (exploratory, declared as such) —** expectancy of voice content vs preceding EEG.

Instrument. Eight wet Ag/AgCl EEG channels (10–20: Fz, Cz, Pz, C3, C4, T7, T8, F7), 3 EMG, 250 Hz+, analysis capped at 30 Hz. Audio envelopes (stimulus, voice microphone, room microphone) and button states are digitised on the same ADS1299 converter as the EEG, so stimulus onset, speech offset and brain response share one sample clock — no post-hoc alignment. Self-built from open hardware (€230–305/kit) or a commercial equivalent for the pilot; validated against a commercial amplifier on the same person before use. Room audio is discarded at ingest outside scripted windows.

Discipline. Full pre-registration on OSF (electrode, window, test, minimum trials, exclusion criteria, primary vs exploratory), blinded analysis with the group key held separately, analysis code frozen and hashed before data collection and tested on simulated data with planted and null effects, controls analysed first as a gate on the rig, trial counts equalised by subsampling, medication, hearing thresholds, vocal level, sleep and caffeine recorded. A guided session runner on TI One Voice enforces impedance and timing gates, shows no live trace, labels button timelines as reports not findings, carries a stop control on every

screen, and returns no individual verdict. A bounded feedback protocol exists for those who opt in: separately consented, three sessions, a reliability gate ($r \geq 0.7$), a three-state vocabulary (consistent with A / with B / not distinguishable) at pre-registered likelihood-ratio thresholds, released only after Stage 1 against frozen reference distributions, delivered in a scheduled conversation.

Open protocol decisions where your input is sought now. Whether to add an explicit resting baseline (4 min eyes open, 4 eyes closed) and a passive oddball block as a seventh block; fixing one sampling rate (250, 500 or 1000 Hz) for the fleet unit; a three-state (pass / marginal / fail) silent-reply quality gate. Each is flagged in the design rather than silently chosen.

Ethics and data. Independent ethical review (the reason for this letter), a data protection impact assessment before recruitment, explicit per-protocol consent naming the unwelcome outcome, a staffed support pathway separate from the research team, EU health-data handling with local processing only, and a commitment to publish the null.

3. Expected results

The fork, decided in advance. Outcome A (the voices behave like inner speech): reduced N100 to the reported voice, a motor build-up in the second before onset, frontal activity leading temporal. Outcome B: full-sized N100, no build-up, temporal leading frontal. All three discriminators fit inside the eight-channel headset, with the stated caveats that the first needs the speech-offset-locked timing of the provocation variant and the second is a consistency check rather than a discriminator. Even Outcome B would not be evidence of transmission, mechanism or source — it would move the question from whether to where.

Primary outcome (Test 1)	Reading	Likelihood and value
Study group shows reduced N100 suppression, like the clinical group	Authorship-tagging mechanism miscalibrated; supports the internal-generation account	Most likely a priori. Converts the standard account from assumed to measured for this population
Study group shows normal suppression, like controls	Tagging intact; the standard misattribution model does not fit this group	Least likely. Would demand an explanation nobody currently has
Study group sits between, or splits into subgroups	The label covers more than one phenomenon	Plausible and arguably most useful: a physiological basis for separating subgroups

Secondary expectations: Test 2 establishes whether reported episodes coincide with a measurable change in auditory-cortex state (attention alone would produce this, so it is necessary, not sufficient). Test 3 yields a calibrated probability that silent replies resemble addressed speech (B) or inner monologue (A); an underpowered "neither" is a likely and reportable outcome on eight channels. Reply-latency distributions from the provocation variant are diagnostic in their own right and have never been measured. For individual feedback, at $d = 0.8$ roughly four in five opted-in participants would receive "not distinguishable" on the Test 1 placement; the design says so before enrolment.

4. Expected impact

- **An empirical gap closed.** The first instrumented study of a population this size, using a validated paradigm, with a result that is publishable whichever way it falls.
- **A methods contribution.** How to study a contested experience in a population that distrusts researchers without a design that can only confirm, and how to give participants bounded individual feedback without manufacturing false certainty — applicable to contested illness generally. This is the contribution that outlives the specific finding.
- **A route back into care.** These are people in serious distress, largely disengaged from clinical services they experience as dismissive. A study that treats them as participants rather than patients to be corrected re-opens contact.
- **Outputs.** Three pre-registered papers: a protocol/methods paper, the primary three-group findings, and — conditional on a defined signature — a distributed replication across a circulating fleet of kits. Derived features (never raw audio) deposited for re-use under the consent obtained.

5. Funding strategy

The base programme is self-funded and does not depend on external money to start. What is not yet funded is academic contribution, including trainee time, and we say so plainly rather than obscure it.

Stage	Cost	Source
0 — Proof of protocol: prototype hardware, timing chain, 6–8 pilot sessions	€2,150	Self-funded by TI One Voice. Ethical review assumed free via a partner institution (commercial review would add ~€6,000).
1 — Decisive study: 10 kits, 75 sessions, independent statistical review	€10,900	Self-funded to the extent required; supplemented by donations and a crowdfunding campaign launched at Stage 0.

2 — Fleet replication: 25 kits, 300 sessions	€22,900	Participant deposit (€150, refundable) and contribution (€50) model recovers ~two-thirds; remainder from donations. A ring-fenced hardship fund keeps ability to pay out of the cohort.
Second instrument for the sub-cortical branch (brainstem response, envelope tracking against silent microphones, otoacoustic emissions): 16 kHz sampling, in-ear and emission probes	~€1,400–2,800 per kit	Out of scope for this study. Pursued only if Outcome B is observed, as a separate design and a separate grant. The current headset is not claimed to reach those stages.
Academic contribution, trainee salary, publication fees	To be defined with partner	Grant applications (EU neurotechnology / digital-rights funding, mental-health and cognitive-liberty foundations). Where the partner institution is the stronger applicant, we co-apply with the partner as PI.

What we ask of a partner: methodological challenge to the protocol before pre-registration; hosting of the ethics review; advice on the statistical plan with a veto over anything that overreaches; help reaching the clinical comparison group; a named methodological co-author. **What we do not ask:** funding, lab space, staff time in volume, equipment, clinical responsibility for participants, or agreement with any interpretation. Hardware, firmware, software, logistics and participant support are carried by TI One Voice and its volunteers. Our position is that the study will run as well as our means allow; additional funding raises its quality — a funded trainee, larger cohorts, independent review — rather than deciding whether it happens.

6. Realistic timeline

Period	Milestones
Sep–Dec 2026	Secure academic partner; protocol challenged and revised; Stage 0 hardware prototyped (three board iterations); timing chain validated by loop-back; DPIA and consent architecture drafted; crowdfunding and grant applications opened.
Jan–Mar 2027	Ethics submission through partner institution; pre-registration drafted and sent for independent statistical review; analysis pipeline built and tested on simulated data; analysis scripts frozen and hashed; support-pathway protocol written and staffed.
Apr–Jun 2027	Pilot (6–8 participants, both groups) on two kits: feasibility of unsupervised two-hour sessions, artifact levels, reply-latency check. Go / no-go on Stage 1. Pre-registration published.
Jul 2027–Mar 2028	Stage 1: ten kits circulating; ~75 sessions across three groups (clinical group recruitment via partner is the pacing item). Controls analysed first as a rig gate.
Apr–Jun 2028	Unblinding, confirmatory analysis, methods and findings papers submitted. Decision on Stage 2 fleet replication, conditional on a defined signature and funding.

Assumes ethics approval within one review cycle; each stage is gated on the one before and can be stopped with a written account of why.

Further reading

- [Can a cheap EEG headset test when people report hearing V2K voices?](#) — full article; [video version](#).
- [Nine minutes inside a session that has never been recorded](#) — participant and administrator screens, and both analysis branches.
- [Full design document](#) — version 3 (~150 pages, attached): paradigms, hardware, statistical plan, the Outcome B branch, the individual-feedback protocol, ethics, recruitment. An appendix, not a prerequisite.

Contact: Stephane van der Aa, stephane@stepvda.com · one.witysk.org